viewed as combining two stages in parallel, where one of the stages happens to be the *identity system* (the output is identical to the input). Using this approach, it can be shown that the "b" coefficients are left unchanged, and the modified "a" coefficients are given by:

EQUATION 33-6
Spectral inversion. The frequency response of a recursive filter can be flipped top-for-bottom by modifying the "a" coefficients according to these equations. The original coefficients are shown in italics, and the modified coefficients in roman. The "b" coefficients are not changed. This method usually provides poor results.

$$\begin{aligned} a_0 &= 1 - a_0 \\ a_1 &= -a_1 - b_1 \\ a_2 &= -a_2 - b_2 \\ a_3 &= -a_3 - b_3 \\ &\vdots \end{aligned}$$

Figure 33-6 shows spectral inversion for two common frequency responses: a low-pass filter, (a), and a notch filter, (c). This results in a high-pass filter, (b), and a band-pass filter, (d), respectively. How do the resulting frequency responses look? The high-pass filter is absolutely terrible! While

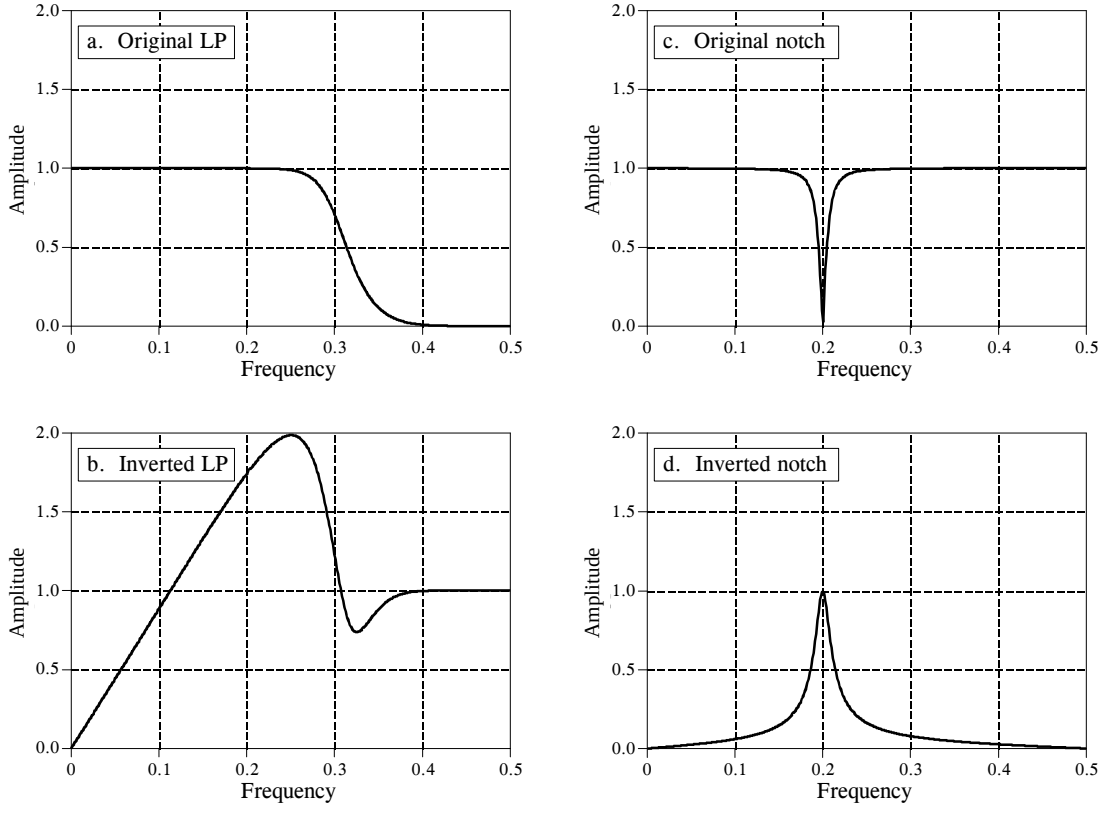

FIGURE 33-6
Examples of spectral inversion. Figure (a) shows the frequency response of a 6 pole low-pass Butterworth filter. Figure (b) shows the corresponding high-pass filter obtained by spectral inversion; its a mess! A more successful case is shown in (c) and (d) where a notch filter is transformed in to a band-pass frequency response.